

Exp No. 5: MIMO System Simulation with BER Analysis Date

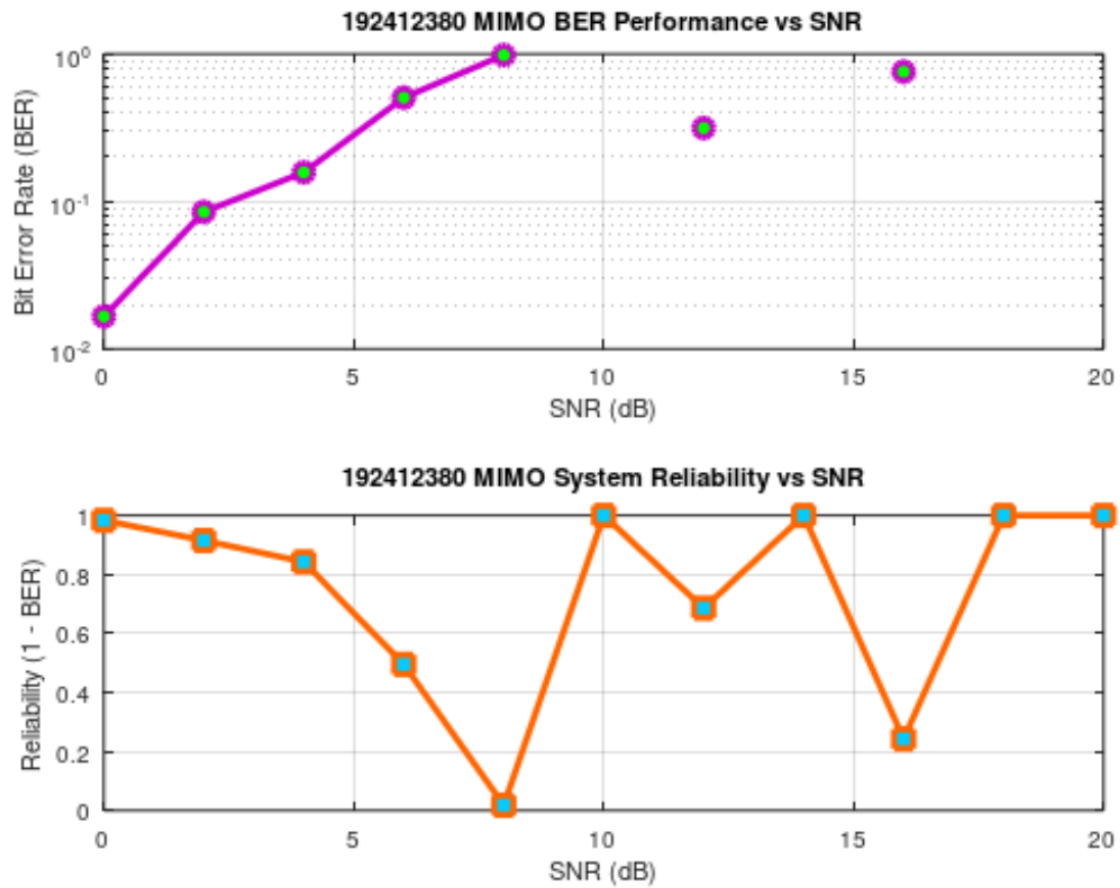
AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective 1: To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.

Program :

```
1  clc;
2  clear;
3  close all;
4
5  % SNR Range
6  SNR_dB = 0:2:20;
7  SNR = 10.^(SNR_dB/10);
8
9  % MIMO Parameters
10 Nt = 2;
11 Nr = 2;
12
13 BER = zeros(1,length(SNR));
14
15 for i = 1:length(SNR)
16
17     % Generate Random Bits
18     bits = randi([0 1],10000,1);
19
20     % BPSK Modulation
21     symbols = 2*bits - 1;
22
23     % MIMO Channel
24     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
25
26     % AWGN Noise
27     noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
28
29     % Transmission (Single Stream)
30     tx = H(1,1)*symbols' + noise(1,:);
31
32     % Detection
33     rx = real(tx) > 0;
34
35     ..
```

OUTPUT :



Objective 2: To analyze the impact of **Signal-to-Noise Ratio (SNR in dB)** on the reliability and quality of MIMO communication.

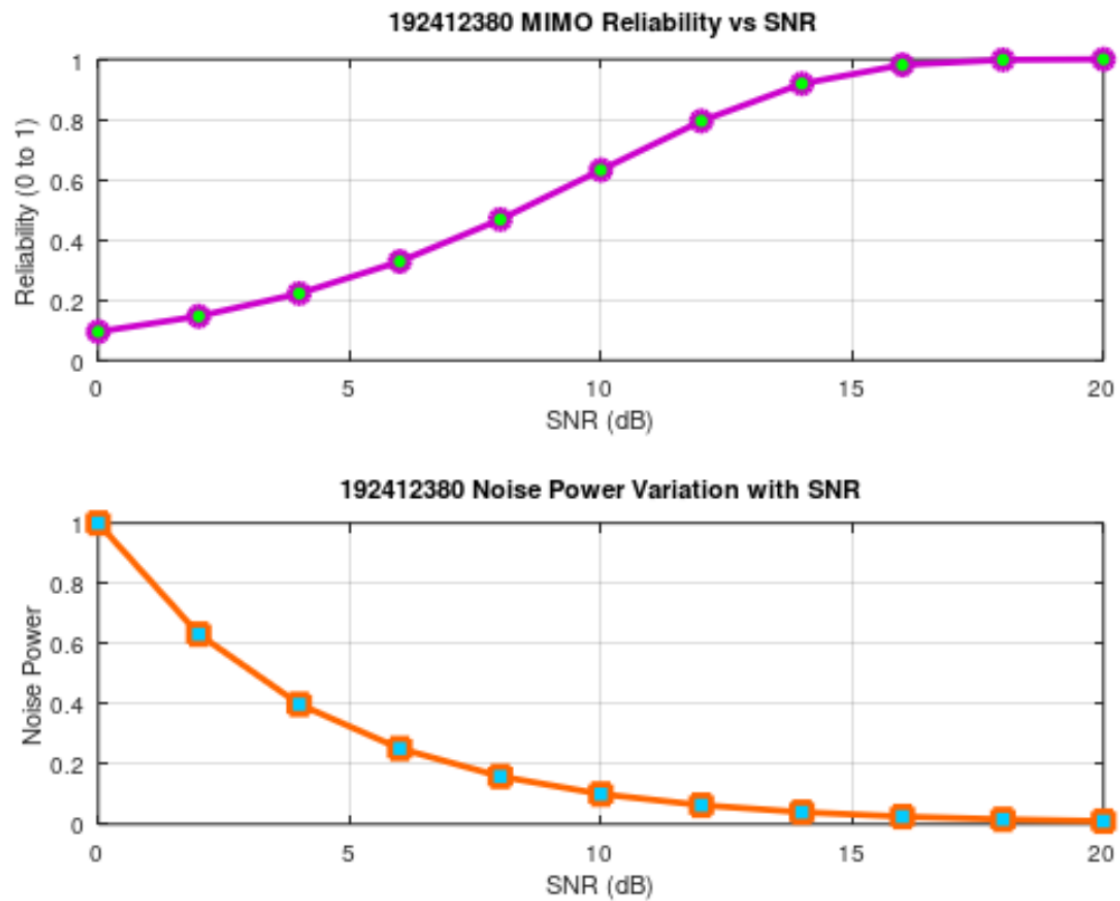
Program

```

1  clc;
2  clear;
3  close all;
4
5  % SNR Range (dB)
6  SNR_dB = 0:2:20;
7  SNR = 10.^(SNR_dB/10);
8
9  % MIMO Parameters
10 Nt = 2;
11 Nr = 2;
12
13 % Initialize Arrays
14 Reliability = zeros(1,length(SNR));
15 NoisePower = zeros(1,length(SNR));
16
17 for i = 1:length(SNR)
18
19     % Simple MIMO Channel
20     H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
21
22     % Noise Power
23     NoisePower(i) = 1/SNR(i);
24
25     % Reliability Model
26     Reliability(i) = 1 - exp(-SNR(i)/10);
27
28 end
29
30 figure;
31
32 %----- Graph 1 -----%
33 subplot(2,1,1);
34

```

OUTPUT :



Objective 3: To compute the Channel Capacity (bps/Hz) and study how MIMO improves spectral efficiency compared to single antenna systems.

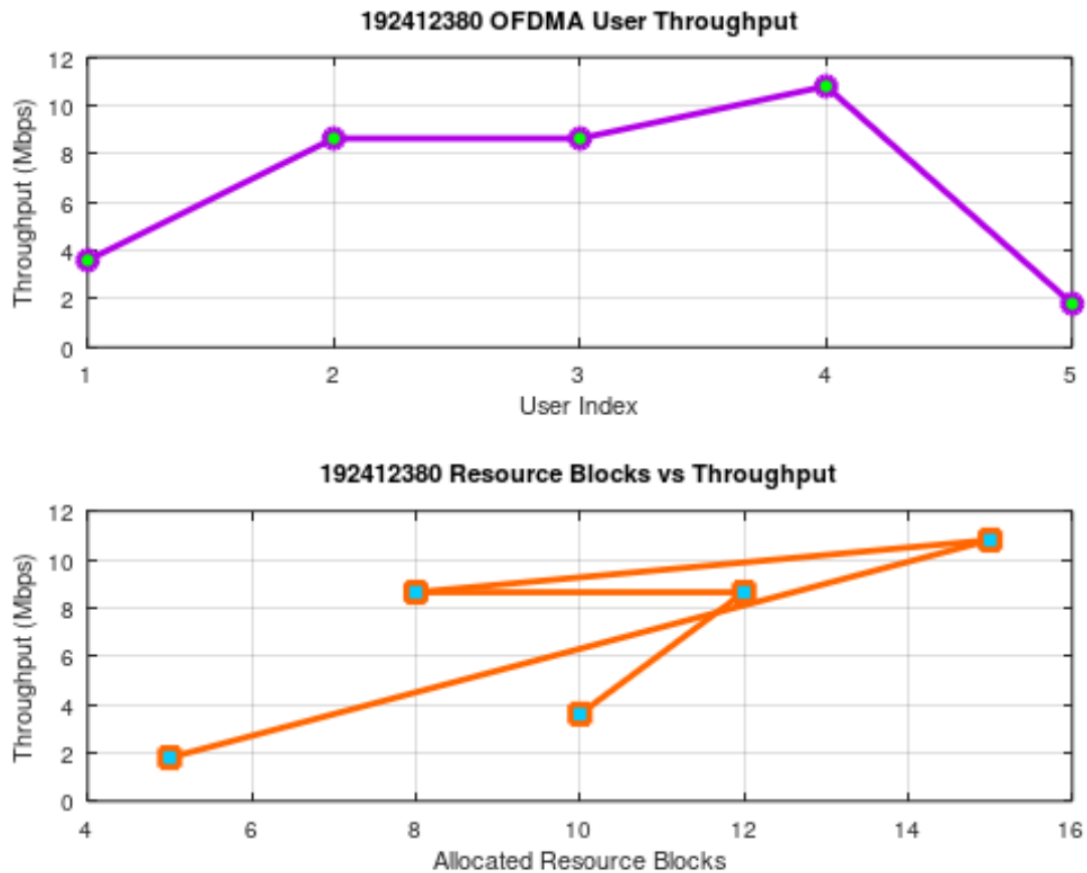
program

```

1  clc;
2  clear;
3  close all;
4
5  % Resource Block Parameters
6  rb_bw = 180e3;      % 180 kHz per Resource Block
7
8  % Users
9  users = 1:5;
10
11 % Allocated Resource Blocks per User
12 alloc_RB = [10 12 8 15 5];
13
14 % Modulation Efficiency (bits/symbol)
15 % QPSK = 2, 16QAM = 4, 64QAM = 6
16 mod_eff = [2 4 6 4 2];
17
18 % Throughput Calculation (Mbps)
19 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
20
21 % Display Results
22 fprintf('User\tAllocated RB\tThroughput (Mbps)\n');
23 fprintf('-----\n');
24
25 for i = 1:length(users)
26     fprintf('%d\t%d\t%.2f\n', users(i), alloc_RB(i), throughput(i));
27 end
28
29 figure;
30
31 %----- Graph 1 -----
32 subplot(2,1,1);
33
34 plot(users, throughput, ...

```

OUTPUT :



Result:

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.
2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.
3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance.

Practical Questions

An OFDMA-based wireless communication system allocates available spectrum in the form of resource blocks (RBs) to multiple users. Using MATLAB, simulate the system and analyze user throughput performance.

1.

To determine the Number of Resource Blocks (RBs) allocated to each user based on system configuration.

2.

To evaluate the impact of RB allocation on User Throughput (Mbps) in the OFDMA system.

3.

To analyze how modulation efficiency affects the overall throughput performance of users.